Paid Is Not Scheduled

Re-engineering the Bambinos.live Student Renewal → Class Scheduling pipeline · AI Program Manager (Operations) take-home · Nov 2026

Prototype + data + outputs: see GitHub repo (README). Analysis run on the provided 5-table dataset, snapshot **as of 2026-11-04 09:00**.

> **One-line thesis.** The pipeline treats the *teacher's form* as the trigger and the *payment* as something to verify. It should be the other way round. The gateway already knows who paid, how much, for which student, class and teacher. For a same-batch renewal (the majority) Steps 2 and 3 add no information, only delay and typos. Remove them, let the ledger drive scheduling, and send humans only the cases that genuinely need judgement.

11 / 12	₹1.27 L	66.7 h	8 of 12
paid renewals NOT scheduled (all >12h)	collected revenue with no classes booked	oldest paid-but-unscheduled (a clean same-batch case)	clear automatically with the new rules (67% straight-through)

A. Root-Cause Problem Definition & Leakage Audit

A1. What the data actually shows (every invoice traced through all 4 tables)

Invoice	Where it is stuck	Hidden failure mode	Root cause
84901	Scheduled in 1.7h	The only one that worked: Sunday payment, teacher fast, Ops fast.	Depends on 3 people being online at the same time.
84902	booking = Pending_Batch_Allocation, **66.7h**	Approved same-batch renewal still waiting for someone to "choose" the batch the child is already in.	Step 4 is manual even when there is nothing to choose.
84903	form Pending, **47.9h**	Perfectly clean case: amount, IDs, batch all match. Sitting in the Ops queue for 2 days.	Ops verifies 100% of rows by hand, so clean ones queue behind messy ones.
84904	form Pending	Mother paid from her own phone/email (`...XXXX77`, outlook vs gmail). A phone-based match fails.	Matching on contact details, not system IDs (the invoice carries student_id/user_id).
84905	booking = **Failed_Batch_Full**	Batch 9205 is 6/6, but one of those 6 *is this child*. A renewing student is refused their own seat.	Capacity check doesn't exclude the renewing student.
84906	No form, 41h	**Orphaned invoice** (GCC parent, ₹18,000). Nobody knows this child paid.	Pipeline only starts when the teacher remembers to fill a form.
84907	form Pending	Teacher gave a 20% "retention scholarship" (₹1,900). No approval trail.	No discount policy enforced at the payment-link stage.
84908	2 forms Pending	Same renewal submitted twice ("browser froze"). Risk of double booking.	No idempotency key on the form.
84909	form Pending	Installment 1 of 2: ₹5,250 captured, but the teacher typed ₹10,500.	Teacher re-types financial data the gateway already has.
84910	No form, 21.7h	Second orphaned invoice (Coach Arvind).	Same as 84906.
84911	form Pending	Payment **Failed**, but the teacher submitted a form with a "UPI processing" screenshot. A tired Ops agent could approve it.	The form allows a screenshot to stand in for gateway proof.

84912	Awaiting_Parent_Batch_Selection	Ops *approved* a batch change to Sat-Sun 11:00, but class 301 has no weekend batch. The parent is asked to pick from options that don't exist.	Approval happens before a feasibility check.
84913	form Pending	Teacher typed invoice **84931** (digits swapped, real one is 84913). Parent name also contains a **Cyrillic "Me"** that looks like Latin "Me". Exact-text matching fails silently.	Free-text keys and no input validation.

A2. Failure-mode taxonomy (beyond this sample)

- **Trigger failures:** teacher forgets, is on leave, or files late (orphaned invoices). Weekend and late-evening payments wait until Monday because Ops is offline. Two of 13 payments here came in on a Sunday.
- **Identity failures:** a different payer phone or email, siblings under one parent account, NRI numbers (+971, +1), look-alike characters, masked PII.
- **Money failures:** partial or installment payments, failed or pending UPI, unapproved discounts, currency mismatches (US/GCC), refunds or chargebacks after scheduling.
- **Data-entry failures:** wrong invoice ID or link, wrong amount, wrong batch, duplicate submissions.
- **Scheduling failures:** the "batch full" logic counts the renewing child, batch changes are approved without a feasible slot, and parents don't answer slot-selection messages (open loop).
- **Continuity failures:** the old term ends before the new bookings exist, so the child misses classes even though the parent paid. This is the failure the parent actually feels.
- **Visibility failures:** no stage timestamps or owner, so nobody can see "paid 3 days ago, still nothing".

A3. Metrics & SLAs that are broken or invisible today

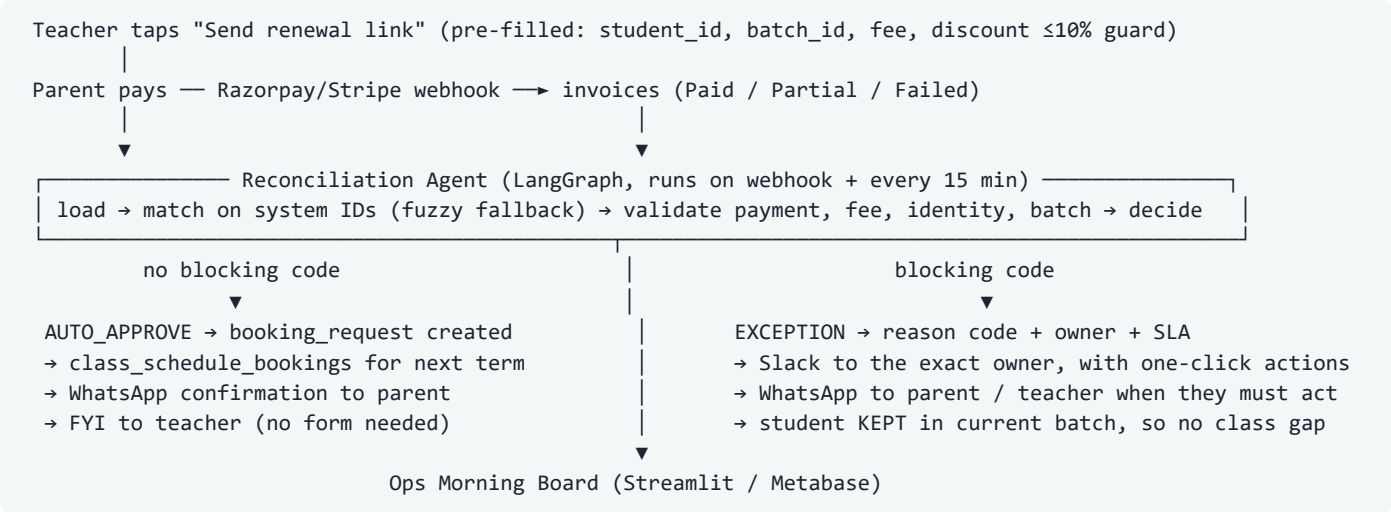
Metric	Definition	Today (sample)	Target SLA
Payment → Schedule TAT	paid_at → first class_schedule_bookings.created_at	1.7h for the one success; the other 11 are still open after 17–67h	Same-batch ≤ 15 min (P95). Exceptions ≤ 24h.
Paid-not-scheduled >12h	Count and ₹ of Paid/Partial invoices with no schedule after 12h	11 · ₹1,26,850	0 for clean cases
Orphaned Invoice Rate	Paid renewals with no teacher form	16.7%	Irrelevant after redesign (form not needed)
Straight-Through Rate	Renewals scheduled with zero human touch	0% (every case is manual)	≥ 70% by Day 60
Form Data Error Rate	Forms with wrong invoice, amount or a duplicate	25%	Form removed for same-batch renewals
Exception Ageing by Owner	Open blocking exceptions × hours × owner	Invisible	Owner-specific SLA (4h / 12h / 24h)
Class-Gap Incidents	Paid students who missed a class because the new term wasn't booked	Invisible	0
False Rejection Rate	Bookings failed for a reason that wasn't true (e.g. false "batch full")	1 of 4 booking requests	0

B. Re-Engineered Workflow — "Zero-Touch" for the Happy Path

B1. Question each step before automating it

Step today	Why it exists	Is that still true?	Redesign
2. Teacher Renewal Form	To tell Ops who paid and which batch	No. The invoice already has student_id, class_id, educator_id and the amount. The form only re-types them, with errors.	**Eliminated** for same-batch renewals. It becomes an optional one-tap "change batch / note" link.
3. Ops manual approval	To check that the payment is real	No. A gateway webhook (payment_id plus Paid status) is stronger proof than a human looking at a link.	**Eliminated** for clean cases. Rules approve, and humans see only exceptions.
4. Batch selection	To pick days and time	Only for batch changes. A same-batch child already has a seat.	**Auto-allocated** to the current batch. A human or the parent picks only on a real change.
Payment link creation	Teacher sends a generic link	The root of the data problems	**Strengthened:** the link is generated per student and batch, pre-tagged, with discount guardrails.

B2. Target flow



Guardrail: every approve/reject decision is a deterministic rule (`src/rules.py`), so it is auditable and testable. The LLM (Gemini) only rewrites message tone. Code checks that every number survives the rewrite, otherwise the template is sent.

B3. Smart exception handling: the right person, not "Ops"

Situation	Rule	Who is alerted	Outcome
Parent paid from a different phone/email (84904)	IDs on the invoice match, so identity is proven	Parent (confirm 2nd number). Ops FYI.	Auto-approved
Teacher typed the wrong invoice ID (84913)	Unique match on student+user+class+amount	Teacher FYI ("we fixed it")	Auto-approved, correction logged
Teacher typed the wrong amount (84909)	Gateway amount wins	Teacher FYI	Data corrected
Installment payment (84909)	Partially_Paid	Ops Finance (Slack, 12h) + parent receipt	Exception
Payment failed, screenshot sent (84911)	Gateway ≠ Paid → never approve	Parent gets a retry link. Teacher told "no action".	Blocked

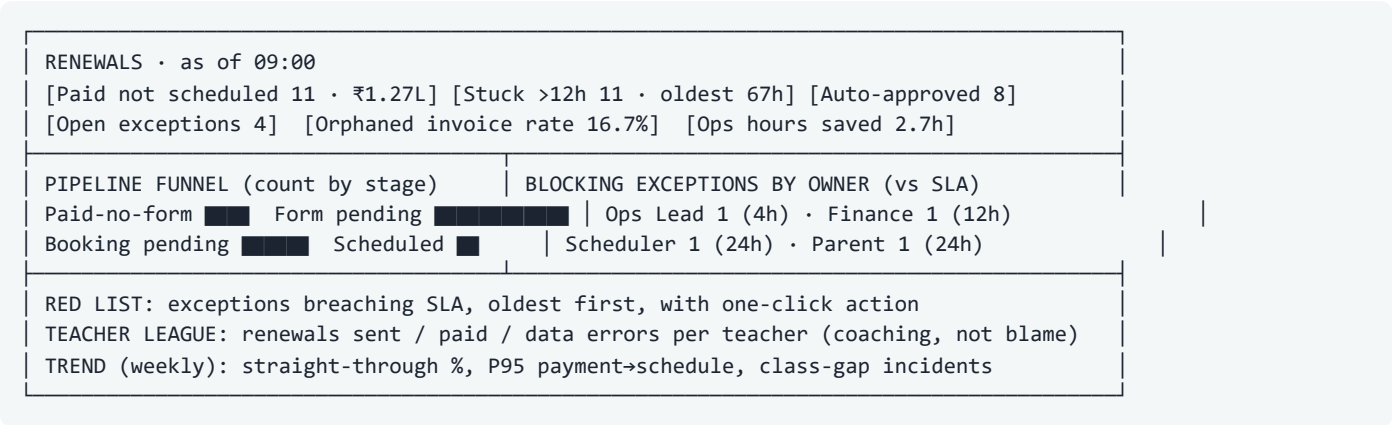
Discount >10% (84907)	Policy limit	Ops Lead (Slack APPROVE/REJECT, 4h)	**Exception**
Batch change to a slot that doesn't exist (84912)	No batch of class 301 on Sat-Sun 11:00	Ops Scheduler (24h) + parent reassurance	**Exception, child stays in current batch**
False "batch full" (84905)	Renewing student already holds a seat	Ops Scheduler FYI + bug ticket	**Auto-approved**
Duplicate form (84908)	Same student+invoice+amount	Teacher FYI	**Earliest kept, duplicate closed**
No teacher form (84906, 84910)	Paid + IDs match roster	Teacher FYI	**Fast-tracked into the same batch**

B4. Prototype result on the provided dataset (Deliverable C, summary)

`python src/pipeline.py` processes all 5 CSVs in a LangGraph StateGraph and writes: **(1)** 8 auto-approved renewals with 192 `class_schedule_bookings` rows ready to insert, **(2)** 13 flagged exception lines with exact reason codes, owner and SLA (4 blocking cases), and **(3)** 22 drafted WhatsApp/Slack messages addressed to Parent, Teacher, Ops Lead, Ops Finance or Ops Scheduler. All 13 invoices were classified correctly against a manual trace.

D. Ops Head Dashboard, SOP & 30-60-90 Rollout

D1. The 9 AM board (built as `dashboard/app.py`)



Hours saved: today each renewal costs about 20 min of manual work (teacher form ~6, Ops verification ~8, allocation ~6) plus ~10 min of chasing per exception. Assuming **~400 renewals a week** and 70% straight-through: 280 × 20 min ≈ **93 h/week saved**. Exceptions arrive pre-diagnosed, so ~10 min saved on each of 120 adds ~20h. Net **≈110 h/week ≈ 2.5–3 FTE** of a 10-person Ops team, redeployed to exception handling and parent experience. Teachers save ~40 h/week of form-filling. (Assumptions to validate in the first 30 days by time-and-motion sampling.)

D2. SOP for the new world

Role	Daily routine
Ops Agent (×6)	Work only the "My exceptions" queue, oldest first. Each card shows the reason code, evidence and the drafted message. Resolve with one click (approve / send message / reassign). Never approve a case that the gateway doesn't show as Paid.
Ops Scheduler (×2)	Handle batch changes, new-batch openings and FALSE_BATCH_FULL tickets. Rule: the child stays in the current batch until the move is booked.
Ops Lead / Finance (×2)	Discount approvals (4h), installment decisions (12h), weekly rule tuning from the exception log.
Teacher (200+)	Send the renewal link from the app. No form. Reply to the WhatsApp nudge only if the parent wants a different slot.
Weekend	The engine runs 24×7, so same-batch renewals need no one. One on-call agent covers blocking exceptions with a 12h weekend SLA.

D3. Change management: rolling out without breaking live classes

- **Shadow mode first.** The engine runs alongside the manual process and writes to a *shadow* table. Ops keeps working as today. Every day we compare engine decisions with human decisions and target ≥98% agreement before switching on.
- **Never delete a seat.** In every mode, an existing student is never removed from the current batch until the new bookings are written. That rules out class gaps by design.
- **Kill switch and rollback.** A single config flag sends everything back to the manual queue. Auto-approvals are reversible for 24h.
- **Teachers:** the message is "you no longer need to fill a form", which removes work, so adoption is easy. Pilot with 20 high-renewal teachers, then onboard in waves through 10-minute WhatsApp voice notes and a pinned FAQ. The Teacher League rewards teachers, it doesn't name and shame them.
- **Ops team:** roles move from "verifier" to "exception specialist". Two Ops champions co-own the rule book, and nobody is made redundant in the first 90 days.

D4. 30-60-90 day plan

Window	Build	Rollout	Exit criteria
Days 0–30 See & shadow	Gateway webhook → engine. Stage timestamps on all 4 tables. Morning Board live. Fix the FALSE_BATCH_FULL capacity bug.	Shadow mode across 100% of renewals. Time-and-motion baseline. Clear the current backlog using the engine output.	Board used daily. ≥98% engine–human agreement. Zero paid-not-scheduled >72h.
Days 31–60 Fast-track	Same-batch auto-approve switched on. Slack one-click actions. WhatsApp templates approved by Meta. Gemini tone layer.	Pilot with 20 teachers, then 100. Form becomes optional for same-batch renewals.	Straight-through ≥60%. P95 same-batch TAT ≤1h. Class-gap incidents = 0.
Days 61–90 Remove & scale	Per-student renewal links with a discount guard. Parent self-serve slot picker that shows only feasible batches. Installment policy automated.	All 200+ teachers. Teacher form retired. Ops reorganised into exception pods.	Straight-through ≥70%. Ops hours on renewals −60%. Parent CSAT on renewal ≥4.5.

Risks & mitigations

- **Wrong auto-approval:** approval only ever happens on gateway Paid + an ID match. Everything is logged and reversible, and the shadow period proves accuracy first.
- **Webhook outage:** a 15-minute polling reconciliation catches up, and the board shows the time of the last successful sync.
- **Bad LLM messages:** templates are the default. The LLM only rewrites tone, and numbers are checked in code.
- **PII:** data stays masked and only IDs go to the LLM. WhatsApp sends use approved templates only.